

Formula sheet

Solutions

$$\begin{aligned}\mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \\ \mathbf{x}[k] &= \mathbf{A}^k\mathbf{x}[0] + \sum_{m=0}^{k-1} \mathbf{A}^{k-1-m}\mathbf{B}\mathbf{u}[m]\end{aligned}$$

Controllability/Observability

$$\begin{aligned}\mathcal{C} &= [\mathbf{B}, \mathbf{AB}, \mathbf{A}^2\mathbf{B}, \dots, \mathbf{A}^{n-1}\mathbf{B}] \\ \mathcal{O} &= \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix}\end{aligned}$$

Realization

$$\begin{aligned}\mathbf{G}(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \\ \mathbf{G}(s) &= \mathbf{G}(\infty) + \mathbf{G}_{sp}(s) \\ d(s) &= s^r + \alpha_1 s^{r-1} + \dots + \alpha_{r-1}s + \alpha_r \\ \mathbf{G}_{sp}(s) &= \frac{1}{d(s)}[\mathbf{N}_1 s^{r-1} + \mathbf{N}_2 s^{r-2} + \dots + \mathbf{N}_{r-1}s + \mathbf{N}_r] \\ \dot{\mathbf{x}} &= \begin{bmatrix} -\alpha_1 \mathbf{I}_p & -\alpha_2 \mathbf{I}_p & \dots & -\alpha_{r-1} \mathbf{I}_p & -\alpha_r \mathbf{I}_p \\ \mathbf{I}_p & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_p & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_p & \mathbf{0} \end{bmatrix} \mathbf{x} + \begin{bmatrix} \mathbf{I}_p \\ \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} \mathbf{u} \\ \mathbf{y} &= [\mathbf{N}_1 \quad \mathbf{N}_2 \quad \dots \quad \mathbf{N}_{r-1} \quad \mathbf{N}_r] \mathbf{x} + \mathbf{G}(\infty) \mathbf{u}\end{aligned}$$

LQR

$$J = \int_0^\infty \mathbf{x}^\top(t) \mathbf{Q} \mathbf{x}(t) + \mathbf{u}^\top(t) \mathbf{R} \mathbf{u}(t) dt$$

$$\mathbf{A}^\top \mathbf{P} + \mathbf{P} \mathbf{A} + \mathbf{Q} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} = \mathbf{0}$$

$$\mathbf{u}(t) = -\mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{x}(t)$$

Lyapunov equation

$$\mathbf{A}^\top \mathbf{M} + \mathbf{M} \mathbf{A} = -\mathbf{N}$$

Kalman filtering (Discrete time)**Process model**

$$\mathbf{x}[k+1] = \mathbf{A}_d \mathbf{x}[k] + \mathbf{B}_d \mathbf{u}[k] + \mathbf{w}[k], \quad \mathbf{y}[k] = \mathbf{C} \mathbf{x}[k] + \mathbf{v}[k]$$

The noise and disturbance are unbiased ($\mathbb{E}[\mathbf{v}[k]] = \mathbf{0}$, $\mathbb{E}[\mathbf{w}[k]] = \mathbf{0}$) and white

$$\mathbb{E}[\mathbf{v}[k] \mathbf{v}[l]^\top] = \delta[k, l] \mathbf{R}_d, \quad \mathbb{E}[\mathbf{w}[k] \mathbf{w}[l]^\top] = \delta[k, l] \mathbf{Q}_d$$

Algorithm Initialize at $\hat{\mathbf{x}}^-[0] = \mathbb{E}[\mathbf{x}(0)]$ and $\mathbf{P}^-[0] = \mathbb{E}[(\mathbf{x}[0] - \hat{\mathbf{x}}^-[0])(\mathbf{x}[0] - \hat{\mathbf{x}}^-[0])^\top]$.
Compute recursively:

$$1. \mathbf{L}[k] = \mathbf{P}^-[k] \mathbf{C}^\top (\mathbf{C} \mathbf{P}^-[k] \mathbf{C}^\top + \mathbf{R}_d)^{-1}$$

$$2. \hat{\mathbf{x}}[k] = \hat{\mathbf{x}}^-[k] + \mathbf{L}[k] (\mathbf{y}[k] - \mathbf{C} \hat{\mathbf{x}}^-[k])$$

$$3. \mathbf{P}[k] = (\mathbb{I} - \mathbf{L}[k] \mathbf{C}) \mathbf{P}^-[k] (\mathbb{I} - \mathbf{L}[k] \mathbf{C})^\top + \mathbf{L}[k] \mathbf{R}_d \mathbf{L}[k]^\top$$

$$4. \hat{\mathbf{x}}^-[k+1] = \mathbf{A}_d \hat{\mathbf{x}}[k] + \mathbf{B}_d \mathbf{u}[k], \quad \mathbf{P}^-[k+1] = \mathbf{A}_d \mathbf{P}[k] \mathbf{A}_d^\top + \mathbf{Q}_d$$

Kalman filtering (Continuous time)

Process model

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{G}\mathbf{w}, \quad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{v}$$

The noise and disturbance are unbiased ($\mathbb{E}[\mathbf{v}(t)] = \mathbf{0}$, $\mathbb{E}[\mathbf{w}(t)] = \mathbf{0}$) and white

$$\mathbb{E}[\mathbf{v}(t)\mathbf{v}(\tau)^\top] = \delta(t - \tau)\mathbf{R}, \quad \mathbb{E}[\mathbf{w}(\tau)\mathbf{w}(t)^\top] = \delta(t - \tau)\mathbf{Q}$$

Optimal gain The Kalman gain is given by $\mathbf{L}(t) = \mathbf{P}(t)\mathbf{C}^\top\mathbf{R}^{-1}$ where

$$\dot{\mathbf{P}} = \mathbf{A}\mathbf{P} + \mathbf{P}\mathbf{A}^\top + \mathbf{G}\mathbf{Q}\mathbf{G}^\top - \mathbf{P}\mathbf{C}^\top\mathbf{R}^{-1}\mathbf{C}\mathbf{P}$$

Set $\dot{\mathbf{P}} = \mathbf{0}$ to find stationary gain.

Stationary processes

Autocorrelation and power spectral density

$$\mathcal{R}_u(\tau) = \mathbb{E}[u(t)u(t + \tau)], \quad \mathcal{S}_u(\omega) = \mathcal{F}\{\mathcal{R}_u(\tau)\}$$

With $y(s) = H(s)w(s)$ where $\mathbb{E}[w(t)] = 0$ and $\mathcal{R}_w(\tau) = \delta(\tau)q$ it holds that

$$\mathcal{S}_y(\omega) = H(j\omega)H(-j\omega)q$$

Laplace transform pairs

$f(t)$	$\Longleftrightarrow$	$F(s)$
$\delta(t)$	$\Longleftrightarrow$	1
1	$\Longleftrightarrow$	$\frac{1}{s}$
e^{-at}	$\Longleftrightarrow$	$\frac{1}{s+a}$
t	$\Longleftrightarrow$	$\frac{1}{s^2}$
t^2	$\Longleftrightarrow$	$\frac{2}{s^3}$
te^{-at}	$\Longleftrightarrow$	$\frac{1}{(s+a)^2}$
$\sin \omega t$	$\Longleftrightarrow$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\Longleftrightarrow$	$\frac{s}{s^2 + \omega^2}$

Solution of second degree equation

$$ax^2 + bx + c = 0 \quad \Longrightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$